

Proposed Solution: TruthLens – A Vision Language Agent Powered Augmented Reality System for Bangla Multimodal Fake News Detection

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Abstract—The rapid growth of online misinformation demands efficient real-time verification systems. This paper proposes TruthLens, a Vision Language Agent (VLA) powered Augmented Reality (AR) framework for Bangla multimodal fake news detection. By integrating Optical Character Recognition (OCR), Vision Language Models (VLMs), Retrieval-Augmented Generation (RAG), and AR visualization, the system performs evidence-based verification and displays credibility information directly within the user’s camera view.

Index Terms—Fake News Detection, Vision Language Model, Vision Language Agent, Bangla NLP, Retrieval-Augmented Generation, Augmented Reality

I. PROPOSED SOLUTION

The proposed framework, named **TruthLens**, introduces an intelligent multimodal fake news detection system capable of analyzing both textual and visual information simultaneously. Unlike conventional text-only classifiers, TruthLens combines OCR, Vision Language Models, Retrieval-Augmented Generation, and a Vision Language Agent to perform evidence-based reasoning before presenting verification results through an Augmented Reality interface.

The workflow begins by capturing images using a smartphone camera from newspapers, online news portals, television broadcasts, or social media posts. OCR extracts Bangla and English text while removing background noise and formatting inconsistencies. The extracted text is forwarded to the multimodal reasoning pipeline.

Next, a Vision Language Model jointly analyzes textual and visual information to understand semantic relationships, detect contextual inconsistencies, and identify image-text mismatches. This enables the system to recognize misleading multimedia content that cannot be detected using textual information alone.

The extracted claim is transformed into semantic embeddings and processed using a Retrieval-Augmented Generation module. Relevant information is retrieved from trusted news organizations, government portals, and fact-checking repositories. These retrieved documents serve as external evidence for claim verification.

A Vision Language Agent functions as the reasoning engine by combining OCR output, image understanding, retrieved evidence, and user queries. Instead of producing only binary

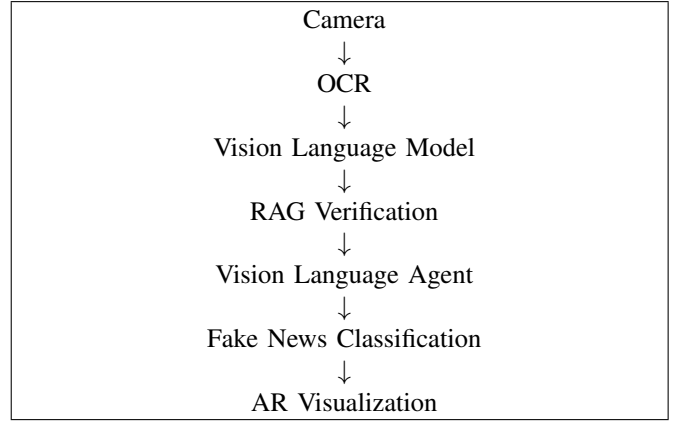


Fig. 1. Overview of the proposed TruthLens framework.

predictions, the agent performs explainable reasoning to determine whether the content is *Real*, *Fake*, *Misleading*, or *Unverified*. The generated explanation improves user trust and transparency.

Finally, the prediction is displayed through an Augmented Reality interface. The system overlays credibility scores, confidence values, supporting evidence, and warning indicators directly onto the camera view, enabling users to verify information without leaving the application.

II. ADVANTAGES

Compared with existing fake news detection systems, TruthLens provides several improvements: (i) multimodal understanding using text and image information, (ii) explainable reasoning through a Vision Language Agent, (iii) retrieval-based evidence verification using trusted sources, (iv) support for Bangla fake news analysis, and (v) real-time AR visualization that enhances user interaction and decision making.

III. EXPECTED OUTCOME

The proposed system is expected to improve fake news detection accuracy while reducing misinformation spread through real-time verification. The integration of multimodal reasoning, explainable AI, and AR visualization offers a practical solution for trustworthy Bangla news verification in mobile environments.